PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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PRODUCT REQUIREMENT DOCUMENT (PRD): RigRefine AI

1. Project Overview

Problem Statement: In heavy industries like Petronas and various O&G SMEs, fresh engineering graduates spend up to 6 months manually sorting, cleaning, and reformatting messy sensor logs from oil rigs. This "Data Janitor" work is a massive waste of high-value talent and creates a productivity bottleneck in decision-making.

Target Domain: Industrial Data Engineering & Decision Intelligence for Heavy Industries (O&G, Manufacturing, Power Plants).

Proposed Solution Summary: RigRefine AI is an autonomous data engineering engine. It uses GLM 5.1 to analyze messy industrial logs, infer complex schemas, and autonomously write/execute Python cleaning scripts. It transforms raw, fragmented data into structured, industry-standard datasets in minutes, featuring a "self-healing" QA loop to ensure 100% accuracy.

2. Background & Business Objective

Background of the Problem: Industrial data is notoriously "dirty." Legacy sensors, varying equipment ages, and lack of standardization lead to logs with inconsistent units, massive gaps, and format drift. Currently, this requires human-in-the-loop manual intervention using Excel or basic scripts that break easily.

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Importance of Solving This Issue: * Talent Retention: Frees junior engineers for high-level analysis rather than manual entry.

- **Speed to Insight:** Reduces data preparation time from months to minutes.
- **Accuracy:** Human error in manual cleaning can lead to catastrophic miscalculations in rig pressure or plant temperature monitoring.

Strategic Fit / Impact: Aligns with the **Industrial 4.0** transition. By moving from "Builder" (writing scripts) to "Curator" (verifying AI-cleaned data), organizations can scale their data operations without linear headcount growth.

3. Product Purpose

3.1. Main Goal of the System: To provide an autonomous, self-healing pipeline that ingest messy industrial logs and outputs "Audit-Ready" clean data and insights.

3.2. Intended Users:

- Data Engineers / Junior Petroleum Engineers
- Plant Operations Managers
- QA Specialists (Industrial Compliance)
- SME Owners (O&G Service Providers)

4. System Functionalities

4.1. Description: The system acts as an autonomous agent. Upon upload, it scans the data, proposes a schema, generates Python code to handle cleaning (imputation, unit conversion, outlier removal), and runs a QA check against industrial bounds.

4.2. Key Functionalities:

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- **Autonomous Schema Inference:** AI identifies what "Column_A" actually represents (e.g., BHP - Bottom Hole Pressure).
- **Self-Healing Code Gen:** If the generated Python cleaning script fails due to a library error or data anomaly, the system catches the error and re-writes the script.
- **Industrial Audit Trail:** Generates a log of every change made to the raw data for regulatory compliance.
- **Before/After Comparison:** Visual diffing of data health before and after the "Refine" process.

4.3. AI Model & Prompt Design

- **4.3.1. Model Selection: GLM 5.1**. Chosen for its superior reasoning in code generation and its ability to handle technical/mathematical contexts common in industrial sensor data.
- **4.3.2. Prompting Strategy: Multi-step Agentic Prompting.** 1. *Analyze:* Scan first 20 rows for schema. 2. *Plan:* List cleaning steps (e.g., "Convert PSI to Bar"). 3. *Execute:* Write and run the script. 4. *Verify:* QA check against the output.
- **4.3.3. Context Handling:** System accepts up to 50MB CSVs. Larger files are chunked; the AI analyzes the first 500 rows to build the "Global Cleaning Logic" which is then applied to the full dataset via the generated script.
- **4.3.4. Fallback Behavior:** If the "Self-Healing" loop fails 3 times, the system flags the specific rows/columns for "Human Intervention" rather than hallucinating a fix.

5. User Stories & Use Cases

User Story: "As a Junior Engineer, I want to upload a year's worth of messy rig logs so that I can get a cleaned dashboard by lunch instead of spending the next quarter in Excel."

Use Cases:

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- **The "Refine" Flow:** User uploads CSV -> AI generates "Cleaning Plan" -> User approves -> AI executes script -> User views "Before/After".
- **The QA Override:** QA Specialist reviews the "Audit Trail" and flags a specific outlier correction as "Invalid," prompting the AI to recalculate.

6. Features Included (Scope)

- Drag-and-drop CSV/Excel uploader.
- Real-time "AI Console" showing code being written.
- Before/After data table comparison.
- Basic Insights Dashboard (Trend lines, Data Health Score).
- Downloadable "Cleaned" CSV and "Audit Log" PDF.

7. Features Not Included (Scope Control)

- Real-time streaming data (Limited to batch file uploads for MVP).
- Direct Write-back to ERP/SAP systems.
- Predictive Maintenance (Focus is on cleaning/engineering, not forecasting).

8. Assumptions & Constraints

- **LLM Cost:** Estimated $0.10 - $0.30 per session due to multiple agentic steps. Decision: Use caching for common schema types (e.g., standard Petronas log headers) to reduce tokens.
- **Technical Constraints:** Code execution happens in a virtualized sandbox. Only standard libraries (Pandas, Numpy, Scipy) are supported.
- **Performance:** Cleaning 100k rows should take < 2 minutes.

9. Risks & Questions

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- **Data Privacy:** How do we ensure sensitive rig coordinates aren't stored in the LLM's history? (Solution: PII Masking before processing).
- **Accuracy Risk:** What if the AI "cleans" a genuine anomaly that was actually a sign of equipment failure? (Solution: QA Specialist must approve "Outlier Removal").
- **Script Stability:** How does the system handle "Mixed Types" (Numbers and Strings) in the same column?

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